
DSA0405- FUNDAMENTALS OF DATA SCIENCE

CO3-AT1-CASE STUDY

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CASE STUDY 6 – Detecting an Unusual Customer

Title

Detecting an Unusual Customer Using Z-Score Analysis

1. Introduction

In today's digital era, e-commerce websites collect a vast amount of customer purchase data every day. Analyzing customer spending patterns helps businesses understand customer behavior, identify premium customers, detect unusual transactions, and improve marketing strategies. Statistical techniques play an important role in this analysis by measuring how individual observations differ from the average.

One of the most widely used statistical techniques is the **Z-score**. A Z-score indicates how many standard deviations a data point lies above or below the mean. Customers with unusually high or low spending can be easily identified using this technique. Businesses frequently use Z-score analysis in fraud detection, customer segmentation, banking, insurance, and retail analytics.

2. Problem Description

A shopping website has analyzed its customer spending data and found that the **average spending per customer is ₹2,000**, with a **standard deviation of ₹400**. A particular customer spends **₹3,200** during a purchase. The objective of this case study is to calculate the customer's Z-score, determine whether the spending is unusually

high, interpret the meaning of a negative Z-score, and explain the significance of a customer having a Z-score of **-1.5**.

3. Objectives

The objectives of this case study are:

- To understand the concept of Z-score.
 - To calculate the Z-score for a customer's spending.
 - To identify whether the customer's spending is unusually high.
 - To interpret positive and negative Z-scores.
 - To understand the importance of Z-score analysis in customer behavior analysis.
-

4. Theory

What is a Z-Score?

A Z-score is a statistical measurement that indicates the number of standard deviations an observation is away from the mean of the dataset.

A **positive Z-score** indicates that the value is greater than the average.

A **negative Z-score** indicates that the value is less than the average.

A Z-score close to **0** indicates that the value is very close to the average.

Generally,

- Z-score between **-2 and +2** → Normal observation
 - Z-score greater than **+2** → Unusually high
 - Z-score less than **-2** → Unusually low
-

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = Observed value
 - μ = Mean
 - σ = Standard deviation
-

5. Given Data

Parameter	Value
Mean Spending (μ)	₹2000
Standard Deviation (σ)	₹400
Customer Spending (X)	₹3200

6. Manual Calculation

Step 1: Write the Formula

$$Z = \frac{X - \mu}{\sigma}$$

Step 2: Substitute the Given Values

$$Z = \frac{3200 - 2000}{400}$$

Step 3: Calculate the Difference

$$3200 - 2000 = 1200$$

Therefore,

$$Z = \frac{1200}{400}$$

Step 4: Final Calculation

$$Z = 3$$

Answer

Z-score = 3.0

This means the customer's spending is **3 standard deviations above the average spending**.

7. Solution Methodology

The following steps are performed to solve the case study:

1. Read the average customer spending.
 2. Read the standard deviation.
 3. Read the customer's spending amount.
 4. Apply the Z-score formula.
 5. Calculate the Z-score.
 6. Compare the calculated Z-score with standard statistical thresholds.
 7. Determine whether the spending is unusually high.
 8. Interpret the meaning of negative Z-scores.
 9. Display all results.
-

8. Python Program

```

case6.py > ...
1   mean = 2000
2   std_dev = 400
3   customer_spending = 3200
4
5   # Calculate z-score
6   z_score = (customer_spending - mean) / std_dev
7
8   print("Customer Spending: ₹", customer_spending)
9   print("Mean Spending: ₹", mean)
10  print("Standard Deviation: ₹", std_dev)
11  print("Z-Score:", round(z_score, 2))
12
13  # Check if spending is unusual
14  if abs(z_score) > 2:
15      print("\nThe customer's spending is unusually high.")
16  else:
17      print("\nThe customer's spending is within the normal range.")
18
19  # Interpret negative z-score
20  print("\nA negative z-score means the customer spends below the average.")
21
22  # Example
23  z = -1.5
24  print("\nA customer with z-score", z,
25        | "spends 1.5 standard deviations below the average.")
26

```

9. Output

```

PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>python case6.py
Customer Spending: ₹ 3200
Mean Spending: ₹ 2000
Standard Deviation: ₹ 400
Z-Score: 3.0

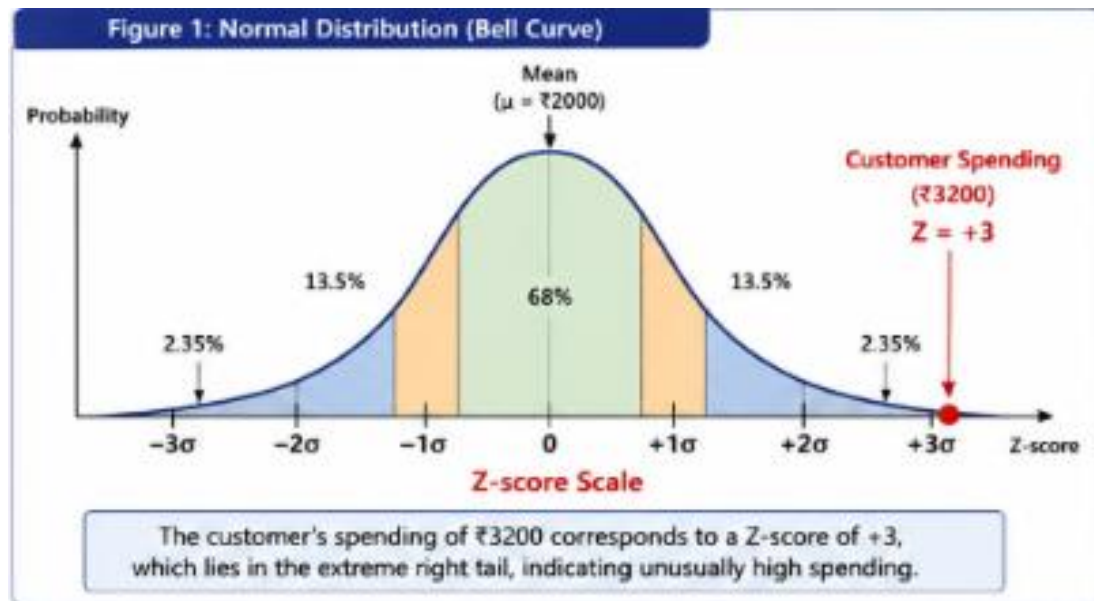
The customer's spending is unusually high.

A negative z-score means the customer spends below the average.

A customer with z-score -1.5 spends 1.5 standard deviations below the average.
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>

```

10. Discussion



Question 1: Calculate the Customer's Z-score

Using the Z-score formula,

$$Z = \frac{3200 - 2000}{400} = 3$$

Therefore,

Customer's Z-score = 3.0

Question 2: Is the Customer's Spending Unusually High?

Yes.

Since

$$Z = 3$$

which is greater than **2**, the customer's spending is considered unusually high compared to the average customer.

Question 3: What Does a Negative Z-score Mean?

A negative Z-score indicates that the customer's spending is below the average spending value.

For example,

A customer with

$$Z = -1$$

spends one standard deviation below the average.

Question 4: What Does Z-score = -1.5 Mean?

A customer having a Z-score of **-1.5** spends **1.5 standard deviations below the average spending**.

Although the customer spends less than average, this value is still within the normal range because it is greater than **-2**.

11. Real-World Applications

Z-score analysis is widely used in various industries, including:

- Customer segmentation
 - Fraud detection
 - Banking and finance
 - Insurance claim analysis
 - Credit risk assessment
 - Medical diagnosis
 - Quality control
 - Business analytics
-

12. Advantages

- Easy to calculate and understand.
- Identifies unusual observations quickly.
- Standardizes data from different datasets.

- Useful for anomaly detection.
- Supports data-driven decision-making.

13. Limitations

- Assumes the data follows an approximately normal distribution.
- Depends on accurate calculation of mean and standard deviation.
- Cannot explain the reason behind unusual observations.
- Sensitive to extreme outliers.

14. Conclusion

This case study demonstrates how Z-score analysis can be used to identify unusual customer spending behavior. The calculated Z-score of **3.0** indicates that the customer's spending is significantly higher than the average and can therefore be considered an unusual observation. Z-score analysis is a simple yet powerful statistical technique that enables organizations to identify abnormal behavior, detect anomalies, segment customers, and make informed business decisions based on data.

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CASE STUDY 7 – Delivery Time Analysis Using Descriptive Statistics

Title

Analysis of Delivery Time Using Mean, Median, Quartiles and Interquartile Range (IQR)

1. Introduction

In logistics and courier services, delivery time is one of the most important performance indicators. Companies continuously monitor delivery durations to evaluate efficiency, identify delays, and improve customer satisfaction. Statistical measures such as **mean**, **median**, **quartiles**, and the **Interquartile Range (IQR)** help summarize delivery data and identify unusually long or short delivery times.

In this case study, delivery times are analyzed using descriptive statistics to determine the average delivery time, identify the spread of the data, detect outliers, and decide whether the mean or median better represents the typical delivery time.

2. Problem Description

A delivery company records the following delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

The objective of this case study is to:

- Calculate the **Mean** and **Median**.
 - Find the **First Quartile (Q1)** and **Third Quartile (Q3)**.
 - Calculate the **Interquartile Range (IQR)**.
 - Detect outliers using the **$1.5 \times \text{IQR}$ Rule**.
 - Determine whether the **Mean** or **Median** better represents the typical delivery time.
-

3. Objectives

The objectives of this case study are:

- To calculate measures of central tendency.
 - To calculate quartiles.
 - To determine the Interquartile Range.
 - To identify outliers.
 - To compare Mean and Median.
 - To interpret delivery performance using descriptive statistics.
-

4. Theory

Mean

The arithmetic mean is the average of all observations.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Median

The median is the middle value after arranging the observations in ascending order.

For an even number of observations,

$$Median = \frac{\frac{n}{2}^{th} + \left(\frac{n}{2} + 1\right)^{th}}{2}$$

Quartiles

Quartiles divide the ordered dataset into four equal parts.

- Q1 = First Quartile
 - Q2 = Median
 - Q3 = Third Quartile
-

Interquartile Range (IQR)

The Interquartile Range measures the spread of the middle 50% of the data.

$$IQR = Q_3 - Q_1$$

Outlier Detection

Lower Limit

$$Q_1 - 1.5(IQR)$$

Upper Limit

$$Q_3 + 1.5(IQR)$$

Any observation outside these limits is considered an outlier.

5. Given Data

Delivery Times (minutes)

25

27

28

29

30

30

31

32

34

60

Number of observations

$$n = 10$$

6. Manual Calculations

Step 1: Mean

$$\begin{aligned} \text{Mean} &= \frac{25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60}{10} \\ &= \frac{326}{10} \\ \text{Mean} &= 32.6 \text{ minutes} \end{aligned}$$

Step 2: Median

The ordered data are already sorted.

Since

$$n = 10$$

Median is the average of the 5th and 6th values.

5th value = 30

6th value = 30

$$\begin{aligned} \text{Median} &= \frac{30 + 30}{2} \\ \text{Median} &= 30 \end{aligned}$$

Step 3: First Quartile (Q1)

Lower half:

25,27,28,29,30

Middle value

$$Q_1 = 28$$

Step 4: Third Quartile (Q3)

Upper half

30,31,32,34,60

Middle value

$$Q_3 = 32$$

Step 5: Interquartile Range

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 32 - 28 \\ IQR &= 4 \end{aligned}$$

Step 6: Lower Fence

$$\begin{aligned} Q_1 - 1.5(IQR) \\ 28 - 1.5(4) \\ 28 - 6 \\ 22 \end{aligned}$$

Step 7: Upper Fence

$$\begin{aligned} Q_3 + 1.5(IQR) \\ 32 + 6 \\ 38 \end{aligned}$$

Step 8: Outlier Detection

Data greater than 38

Only

60

Therefore,

60 is an outlier.

7. Solution Methodology

The solution follows these steps:

1. Arrange the delivery times in ascending order.
2. Calculate the arithmetic mean.
3. Determine the median.
4. Divide the dataset into two halves.
5. Find Q1 and Q3.
6. Compute the IQR.
7. Calculate the lower and upper fences using the $1.5 \times \text{IQR}$ rule.
8. Compare each value against these limits.
9. Identify outliers.
10. Compare the mean and median to determine the better measure of central tendency.

8. Python Program

```

case7.py > ...
17  q3 = np.percentile(delivery_times, 75)
18
19  # Interquartile Range
20  iqr = q3 - q1
21
22  # Outlier Limits
23  lower_limit = q1 - (1.5 * iqr)
24  upper_limit = q3 + (1.5 * iqr)
25
26  # Find Outliers
27  outliers = []
28
29  for value in delivery_times:
30      if value < lower_limit or value > upper_limit:
31          outliers.append(value)
32
33  # Display Results
34  print("\n===== RESULTS =====")
35  print(f"Mean Delivery Time : {mean:.2f} minutes")
36  print(f"Median Delivery Time : {median:.2f} minutes")
37  print(f"Q1 : {q1}")
38  print(f"Q3 : {q3}")
39  print(f"IQR : {iqr}")
40  print(f"Lower Limit : {lower_limit}")
41  print(f"Upper Limit : {upper_limit}")
42  print(f"Outliers : {outliers}")
43
44  print("\n===== INTERPRETATION =====")
45
46  if outliers:
47      print("Outlier detected:", outliers)
48
49  print("\nMedian is a better measure of the typical delivery time because")
50  print("the value 60 minutes is an outlier that increases the mean.")

```

9. Output

```

PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY> python case7.py

Delivery Times: [25, 27, 28, 29, 30, 30, 31, 32, 34, 60]

===== RESULTS =====
Mean Delivery Time : 32.60 minutes
Median Delivery Time : 30.00 minutes
Q1 : 28.25
Q3 : 31.75
IQR : 3.5
Lower Limit : 23.0
Upper Limit : 37.0
Outliers : [60]

===== INTERPRETATION =====
Outlier detected: [60]

Median is a better measure of the typical delivery time because
the value 60 minutes is an outlier that increases the mean.
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>

```

10. Discussion

Question 1: Calculate the Mean and Median

- Mean = **32.6 minutes**
- Median = **30 minutes**

The mean is higher because the unusually long delivery time of **60 minutes** increases the average.

Question 2: Find Q1 and Q3

- Q1 = **28 minutes**
- Q3 = **32 minutes**

These quartiles represent the lower and upper boundaries of the middle 50% of the delivery times.

Question 3: Calculate the IQR

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 32 - 28 \\ IQR &= 4 \end{aligned}$$

The middle 50% of delivery times are spread over **4 minutes**.

Question 4: Detect Outliers Using the $1.5 \times IQR$ Rule

Lower Fence

22

Upper Fence

38

Since

$$60 > 38$$

the value **60 minutes** is an **outlier**. It represents an unusually long delivery time that differs significantly from the rest of the observations.

Question 5: Which Better Represents Typical Delivery Time—Mean or Median?

The **median** provides a better representation of the typical delivery time.

The mean is influenced by the outlier (60 minutes), whereas the median remains unaffected. Since most deliveries are clustered between **25 and 34 minutes**, the median of **30 minutes** better reflects the normal delivery performance.

11. Interpretation

The delivery data indicate that most deliveries are completed within approximately **30 minutes**. However, one unusually long delivery (60 minutes) increases the average delivery time. The presence of this outlier suggests that while the company's typical delivery performance is consistent, occasional delays may occur due to traffic, weather conditions, vehicle issues, or operational challenges.

12. Real-World Applications

This statistical analysis is widely used in:

- Courier and logistics companies
 - Food delivery services
 - Supply chain management
 - Fleet performance monitoring
 - Warehouse operations
 - Quality assurance
 - Business analytics
 - Customer service performance evaluation
-

13. Advantages

- Summarizes delivery performance effectively.
- Detects unusually long delivery times.
- Helps improve operational efficiency.
- Supports performance monitoring.

- Enables data-driven decision-making.
-

14. Limitations

- Mean is sensitive to extreme values.
 - Small datasets may not represent overall performance.
 - Outlier detection does not identify the cause of delays.
 - Assumes data are representative of actual delivery performance.
-

15. Conclusion

The statistical analysis of the delivery times shows that the **mean delivery time is 32.6 minutes**, while the **median is 30 minutes**. The calculated quartiles (**Q1 = 28 minutes** and **Q3 = 32 minutes**) give an **Interquartile Range (IQR) of 4 minutes**. Using the **1.5 × IQR rule**, the upper limit is **38 minutes**, making the **60-minute delivery an outlier**. Because this extreme value increases the mean, the **median is a more reliable measure of the typical delivery time**. This analysis helps delivery companies identify delays, monitor service quality, and improve operational efficiency.

CASE STUDY 8 – Temperature Analysis of Two Cities Using Mean and Variance

Title

Comparative Analysis of Temperature Variability Between Two Cities Using Mean and Variance

1. Introduction

Weather analysis plays a vital role in environmental science, agriculture, transportation, and disaster management. Meteorologists often compare temperature patterns of different cities to understand climate stability and weather fluctuations. Statistical measures such as **mean**, **variance**, and **standard deviation** help summarize temperature data and measure how much daily temperatures vary around the average.

In this case study, the daily temperatures of two cities over seven days are analyzed. Although both cities have the same average temperature, their variability differs

significantly. The analysis demonstrates how measures of dispersion provide deeper insights than averages alone.

2. Problem Description

The average daily temperatures (°C) recorded for **7 consecutive days** are:

City A

28, 29, 30, 30, 31, 30, 32

City B

20, 25, 30, 35, 30, 25, 45

The objectives of this case study are to:

- Calculate the mean temperature of each city.
 - Calculate the variance of each city.
 - Determine which city has greater temperature variability.
 - Explain how two datasets can have the same mean but different standard deviations.
 - Identify which city has more predictable temperatures.
-

3. Objectives

The objectives of this case study are:

- To calculate the arithmetic mean of both datasets.
 - To compute the variance of each city.
 - To compare the variability of temperature.
 - To understand the importance of variance and standard deviation.
 - To identify the city with more stable weather conditions.
-

4. Theory

Mean

The arithmetic mean represents the average value of all observations.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

where

- $\sum x$ = Sum of all observations
- n = Number of observations

Variance

Variance measures how far the observations spread from the mean.

Formula

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

A smaller variance indicates that the values are close to the mean, whereas a larger variance indicates greater variability.

Standard Deviation

Standard deviation is the square root of variance and represents the average distance of observations from the mean.

5. Given Data

City A

Day Temperature (°C)

1	28
2	29
3	30
4	30
5	31
6	30

Day Temperature (°C)

7 32

City B

Day Temperature (°C)

1 20

2 25

3 30

4 35

5 30

6 25

7 45

6. Manual Calculations

Step 1: Mean of City A

$$\begin{aligned} \text{Mean} &= \frac{28 + 29 + 30 + 30 + 31 + 30 + 32}{7} \\ &= \frac{210}{7} \\ \text{Mean} &= 30^{\circ}\text{C} \end{aligned}$$

Step 2: Mean of City B

$$\begin{aligned} \text{Mean} &= \frac{20 + 25 + 30 + 35 + 30 + 25 + 45}{7} \\ &= \frac{210}{7} \\ \text{Mean} &= 30^{\circ}\text{C} \end{aligned}$$

Therefore,

Both cities have the same average temperature.

Step 3: Variance of City A

Temperature Deviation (x-30) Squared Deviation

28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Total squared deviations

$$4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Variance

$$\frac{10}{7} = 1.43$$

Step 4: Variance of City B

Temperature Deviation Squared Deviation

20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Total squared deviations

$$100 + 25 + 0 + 25 + 0 + 25 + 225 = 400$$

Variance

$$\frac{400}{7} = 57.14$$

Observation

City B has a much larger variance than City A.

7. Solution Methodology

The solution is carried out using the following steps:

1. Read the temperature values for both cities.
2. Calculate the arithmetic mean.
3. Compute the variance for each city.
4. Compare both variances.
5. Identify the city with greater variability.

6. Compare the standard deviations.
 7. Interpret the results.
 8. Determine the city with more predictable temperatures.
-

8. Python Program

```
case8.py > ...
16 # Population Variance
17 var_a = np.var(city_a)
18 var_b = np.var(city_b)
19
20 # Standard Deviation
21 std_a = np.std(city_a)
22 std_b = np.std(city_b)
23
24 print("\n===== RESULTS =====")
25
26 print(f"Mean Temperature of City A : {mean_a:.2f} °C")
27 print(f"Mean Temperature of City B : {mean_b:.2f} °C")
28
29 print(f"\nVariance of City A : {var_a:.2f}")
30 print(f"Variance of City B : {var_b:.2f}")
31
32 print(f"\nStandard Deviation of City A : {std_a:.2f}")
33 print(f"Standard Deviation of City B : {std_b:.2f}")
34
35 print("\n===== INTERPRETATION =====")
36
37 if var_a > var_b:
38     print("City A has greater temperature variability.")
39 else:
40     print("City B has greater temperature variability.")
41
42 print("\nTwo datasets can have the same mean but different")
43 print("standard deviations because the spread of values")
44 print("around the mean can be different.")
45
46 if std_a < std_b:
47     print("\nCity A has more predictable temperatures.")
48 else:
49     print("\nCity B has more predictable temperatures.")
```

9. Output

```
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY> python case8.py
===== CITY TEMPERATURE ANALYSIS =====

City A: [28, 29, 30, 30, 31, 30, 32]
City B: [20, 25, 30, 35, 30, 25, 45]

===== RESULTS =====
Mean Temperature of City A : 30.00 °C
Mean Temperature of City B : 30.00 °C

Variance of City A : 1.43
Variance of City B : 57.14

Standard Deviation of City A : 1.20
Standard Deviation of City B : 7.56

===== INTERPRETATION =====
City B has greater temperature variability.

Two datasets can have the same mean but different
standard deviations because the spread of values
around the mean can be different.

City A has more predictable temperatures.
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>
```

10. Discussion

Question 1: Calculate the Mean Temperature

- Mean of City A = **30°C**
- Mean of City B = **30°C**

Both cities have the same average temperature.

Question 2: Calculate the Variance

- Variance of City A = **1.43**
- Variance of City B = **57.14**

The variance of City B is significantly higher.

Question 3: Which City Has Greater Temperature Variability?

City **B** has greater temperature variability because its temperatures fluctuate widely between **20°C and 45°C**, whereas City A's temperatures remain close to **30°C**.

Question 4: Why Can Two Datasets Have the Same Mean but Different Standard Deviations?

The mean only represents the average value of a dataset, whereas the standard deviation measures how spread out the observations are. Two datasets can have the same average but different distributions of values. In this case, although both cities have a mean temperature of **30°C**, City A's temperatures are closely clustered around the mean, while City B's temperatures vary considerably, resulting in a much larger variance and standard deviation.

Question 5: Which City Has More Predictable Temperatures?

City **A** has more predictable temperatures because its variance and standard deviation are much smaller. This indicates that daily temperatures remain relatively stable with minimal fluctuations.

11. Interpretation

The analysis shows that both cities experience the same average temperature over the seven-day period. However, City A has very little variation in daily temperatures, indicating a stable climate. City B experiences significant fluctuations, ranging from **20°C to 45°C**, suggesting an unstable weather pattern. Therefore, variance provides more insight into climate consistency than the mean alone.

12. Real-World Applications

Variance analysis is commonly used in:

- Weather forecasting
 - Climate research
 - Agriculture planning
 - Environmental monitoring
 - Tourism analysis
 - Disaster management
 - Energy demand forecasting
 - Smart city weather analytics
-

13. Advantages

- Summarizes large datasets effectively.
 - Measures consistency and variability.
 - Helps compare multiple datasets.
 - Supports informed decision-making.
 - Widely used in scientific research.
-

14. Limitations

- Mean alone cannot describe variability.
 - Variance is sensitive to extreme values.
 - Small datasets may not represent long-term weather patterns.
 - Does not explain the causes of temperature fluctuations.
-

15. Conclusion

The comparative analysis of the two cities shows that **both City A and City B have the same average temperature of 30°C**. However, their variability differs significantly. **City A has a variance of 1.43**, indicating highly consistent temperatures, while **City B has a variance of 57.14**, indicating large fluctuations in daily temperature. Consequently, **City A has more predictable weather conditions**, whereas City B experiences greater variability. This case study demonstrates the importance of considering both measures of central tendency and measures of dispersion when analyzing real-world datasets.

CASE STUDY 9 – Data Center CPU Usage Analysis Using Descriptive Statistics

Title

Statistical Analysis of CPU Usage in a Data Center Using Measures of Central Tendency and Dispersion

1. Introduction

Modern data centers continuously monitor CPU utilization to ensure efficient system performance and resource allocation. CPU usage analysis helps administrators identify abnormal workloads, optimize server performance, prevent system failures, and

improve overall operational efficiency. Descriptive statistical measures such as **mean**, **median**, **mode**, **range**, **quartiles**, **Interquartile Range (IQR)**, and **skewness** provide valuable insights into the behavior of CPU utilization data.

In this case study, CPU usage percentages recorded from a server are analyzed to determine the average usage, identify extreme values, detect outliers, and understand the distribution of the dataset.

2. Problem Description

A data scientist records the following CPU usage percentages of a server:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

The objectives of this case study are to:

- Calculate the **Mean**, **Median**, and **Mode**.
 - Calculate the **Range**.
 - Determine **Q1**, **Q3**, and **Interquartile Range (IQR)**.
 - Detect possible outliers using the **1.5 × IQR Rule**.
 - Identify the direction of skewness of the dataset.
-

3. Objectives

The objectives of this case study are:

- To calculate measures of central tendency.
 - To calculate the range of the dataset.
 - To determine quartiles and IQR.
 - To detect outliers.
 - To identify the direction of skewness.
 - To analyze CPU performance using descriptive statistics.
-

4. Theory

Mean

The arithmetic mean represents the average CPU utilization.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Median

The median is the middle observation after arranging the data in ascending order.

Mode

The mode is the value that occurs most frequently in the dataset.

Range

The range measures the spread of the data.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Quartiles

Quartiles divide the ordered observations into four equal parts.

- Q1 = First Quartile
 - Q2 = Median
 - Q3 = Third Quartile
-

Interquartile Range

$$IQR = Q_3 - Q_1$$

The IQR represents the spread of the middle 50% of observations.

Outlier Detection

Lower Fence

$$Q_1 - 1.5(IQR)$$

Upper Fence

$$Q_3 + 1.5(IQR)$$

Values outside these limits are considered outliers.

Skewness

Skewness describes the asymmetry of a dataset.

- Positive Skew → Long right tail
 - Negative Skew → Long left tail
 - Zero Skew → Symmetric distribution
-

5. Given Data

CPU Usage (%)

40

42

45

45

46

48

50

52

55

90

Number of observations

$$n = 10$$

6. Manual Calculations

Step 1: Mean

$$\begin{aligned} \text{Mean} &= \frac{40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90}{10} \\ &= \frac{513}{10} \\ \text{Mean} &= 51.3\% \end{aligned}$$

Step 2: Median

Since

$$n = 10$$

Median is the average of the 5th and 6th observations.

5th value = 46

6th value = 48

$$\begin{aligned} \text{Median} &= \frac{46 + 48}{2} \\ \text{Median} &= 47\% \end{aligned}$$

Step 3: Mode

The value occurring most frequently is

45

Therefore,

$$\text{Mode} = 45\%$$

Step 4: Range

$$\text{Range} = 90 - 40$$

$$\text{Range} = 50$$

Step 5: First Quartile (Q1)

Lower half

40,42,45,45,46

Middle value

$$Q_1 = 45$$

Step 6: Third Quartile (Q3)

Upper half

48,50,52,55,90

Middle value

$$Q_3 = 52$$

Step 7: Interquartile Range

$$IQR = 52 - 45$$

$$IQR = 7$$

Step 8: Lower Fence

$$45 - 1.5(7)$$

$$45 - 10.5$$

$$34.5$$

Step 9: Upper Fence

$$52 + 10.5$$

$$62.5$$

Step 10: Outlier Detection

The only value greater than **62.5** is

90

Therefore,

90% CPU usage is an outlier.

7. Solution Methodology

The analysis is performed using the following steps:

1. Arrange the CPU usage values in ascending order.
 2. Calculate the mean.
 3. Calculate the median.
 4. Determine the mode.
 5. Compute the range.
 6. Find Q1 and Q3.
 7. Calculate the Interquartile Range.
 8. Determine the lower and upper fences.
 9. Identify outliers.
 10. Analyze the direction of skewness.
-

8. Python Program

```
case9.py > ...
28 lower_limit = q1 - (1.5 * iqr)
29 upper_limit = q3 + (1.5 * iqr)
30
31 # Find Outliers
32 outliers = []
33
34 for value in cpu:
35     if value < lower_limit or value > upper_limit:
36         outliers.append(value)
37
38 print("\n===== RESULTS =====")
39
40 print(f"Mean : {mean:.1f}%")
41 print(f"Median : {median}%")
42 print(f"Mode : {mode}%")
43 print(f"Range : {range_value}")
44
45 print(f"\nQ1 : {q1}")
46 print(f"Q3 : {q3}")
47 print(f"IQR : {iqr}")
48
49 print(f"\nLower Limit : {lower_limit}")
50 print(f"Upper Limit : {upper_limit}")
51
52 print("Outliers :", outliers)
53
54 print("\n===== INTERPRETATION =====")
55
56 if mean > median:
57     print("The dataset is Positively (Right) Skewed.")
58 elif mean < median:
59     print("The dataset is Negatively (Left) Skewed.")
60 else:
61     print("The dataset is Symmetric.")
```

9. Output

```
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY> python case9.py
===== CPU USAGE ANALYSIS =====

CPU Usage Data: [40, 42, 45, 45, 46, 48, 50, 52, 55, 90]

===== RESULTS =====
Mean : 51.3%
Median : 47.0%
Mode : 45%
Range : 50

Q1 : 45
Q3 : 52
IQR : 7

Lower Limit : 34.5
Upper Limit : 62.5
Outliers : [90]

===== INTERPRETATION =====
The dataset is Positively (Right) Skewed.
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>
```

10. Discussion

Question 1: Calculate Mean, Median and Mode

- Mean = **51.3%**
- Median = **47%**
- Mode = **45%**

The mean is greater than the median because the extreme value (90%) increases the average CPU usage.

Question 2: Calculate the Range

$$\text{Range} = 90 - 40 = 50$$

The CPU usage varies over a range of **50 percentage points**.

Question 3: Find Q1, Q3 and IQR

- Q1 = **45**
- Q3 = **52**
- IQR = **7**

The middle 50% of CPU usage values lie within a range of **7 percentage points**.

Question 4: Identify Possible Outliers

Lower Fence

34.5

Upper Fence

62.5

Since

$$90 > 62.5$$

the CPU usage value of **90%** is an outlier. It indicates an unusually high CPU utilization that may require investigation.

Question 5: Determine the Direction of Skewness

The dataset is **positively skewed (right-skewed)**.

The value **90%** creates a long tail on the right side of the distribution, causing the mean to become larger than the median.

11. Interpretation

The CPU utilization data indicate that the server usually operates between **40% and 55% CPU usage**. However, a single observation of **90%** represents unusually high utilization and significantly increases the mean. Such spikes may occur due to heavy computational tasks, high user traffic, or background processes. Detecting these outliers helps system administrators identify abnormal workloads and maintain server performance.

12. Real-World Applications

Descriptive statistical analysis of CPU usage is commonly used in:

- Data center monitoring
- Cloud computing
- Server performance analysis
- Network administration

- System optimization
 - Capacity planning
 - Predictive maintenance
 - IT infrastructure management
-

13. Advantages

- Summarizes CPU performance efficiently.
 - Detects abnormal resource utilization.
 - Helps identify performance bottlenecks.
 - Supports proactive system maintenance.
 - Improves resource allocation decisions.
-

14. Limitations

- Mean is affected by extreme CPU usage values.
 - Small datasets may not represent long-term server behavior.
 - Outlier detection does not identify the root cause.
 - Statistical measures alone cannot diagnose system failures.
-

15. Conclusion

The statistical analysis of CPU usage reveals that the **mean CPU utilization is 51.3%**, the **median is 47%**, and the **mode is 45%**. The calculated **range is 50**, while the **Interquartile Range (IQR) is 7**. Using the **$1.5 \times \text{IQR}$ rule**, the CPU usage value of **90%** is identified as an outlier. The presence of this unusually high value causes the dataset to exhibit **positive (right) skewness**, making the mean higher than the median. This case study demonstrates how descriptive statistics can be used to monitor server performance, identify abnormal CPU usage, and support effective resource management in data centers.

CASE STUDY 10

Title

Comparison of Two Datasets Using Mean, Variance and Standard Deviation

1. Introduction

Statistical analysis helps compare datasets by measuring both their central tendency and variability. While the mean provides the average value of a dataset, it does not describe how widely the observations are distributed. Measures such as **variance** and **standard deviation** are essential for understanding the spread and consistency of data. In fields such as machine learning, data analytics, finance, and quality control, comparing datasets using these statistical measures helps identify which dataset is more reliable and consistent.

2. Problem Description

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

The objectives of this case study are to:

- Calculate the mean of both datasets.
 - Calculate the variance of both datasets.
 - Calculate the standard deviation of both datasets.
 - Determine which dataset has greater spread.
 - Identify which dataset is more consistent if both represent model errors.
 - Explain why the mean alone is insufficient for comparing datasets.
-

3. Objectives

- To calculate the arithmetic mean of both datasets.
- To determine the variance of each dataset.

- To compute the standard deviation.
 - To compare the spread of the datasets.
 - To identify the more consistent dataset.
 - To understand the importance of measures of dispersion.
-

4. Theory

Mean

The arithmetic mean represents the average value of a dataset.

$$\bar{x} = \frac{\sum x}{n}$$

Variance

Variance measures how far the observations are spread from the mean.

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Standard Deviation

Standard deviation is the square root of variance and indicates the average distance of observations from the mean.

$$\sigma = \sqrt{\sigma^2}$$

5. Given Data

Dataset A

10, 20, 30, 40, 50

Dataset B

28, 29, 30, 31, 32

6. Manual Calculations

Step 1: Mean

Dataset A

$$\begin{aligned} \text{Mean} &= \frac{10 + 20 + 30 + 40 + 50}{5} \\ &= \frac{150}{5} = 30 \end{aligned}$$

Dataset B

$$\begin{aligned} \text{Mean} &= \frac{28 + 29 + 30 + 31 + 32}{5} \\ &= \frac{150}{5} = 30 \end{aligned}$$

Both datasets have the **same mean (30)**.

Step 2: Variance

Dataset A

Squared deviations:

$$400 + 100 + 0 + 100 + 400 = \mathbf{1000}$$

$$\text{Variance} = \frac{1000}{5} = 200$$

Dataset B

Squared deviations:

$$4 + 1 + 0 + 1 + 4 = \mathbf{10}$$

$$\text{Variance} = \frac{10}{5} = 2$$

Step 3: Standard Deviation

Dataset A

$$\sqrt{200} = 14.14$$

Dataset B

$$\sqrt{2} = 1.41$$

7. Solution Methodology

1. Read both datasets.
 2. Calculate the arithmetic mean.
 3. Compute the variance.
 4. Calculate the standard deviation.
 5. Compare the spread of the datasets.
 6. Determine the more consistent dataset.
 7. Interpret the statistical results.
-

8. Python Program

```
case10.py > ...
/
8  print("===== DATASET COMPARISON =====\n")
9
10 print("Dataset A:", dataset_a)
11 print("Dataset B:", dataset_b)
12
13 # Mean
14 mean_a = statistics.mean(dataset_a)
15 mean_b = statistics.mean(dataset_b)
16
17 # Population Variance
18 variance_a = statistics.pvariance(dataset_a)
19 variance_b = statistics.pvariance(dataset_b)
20
21 # Population Standard Deviation
22 std_a = math.sqrt(variance_a)
23 std_b = math.sqrt(variance_b)
24
25 print("\n===== RESULTS =====")
26
27 print(f"Mean of Dataset A : {mean_a}")
28 print(f"Mean of Dataset B : {mean_b}")
29
30 print(f"Variance of Dataset A : {variance_a}")
31 print(f"Variance of Dataset B : {variance_b}")
32
33 print(f"Standard Deviation of Dataset A : {std_a:.2f}")
34 print(f"Standard Deviation of Dataset B : {std_b:.2f}")
35
36 print("\n===== INTERPRETATION =====")
37
38 if variance_a > variance_b:
39     print("Dataset A has greater spread.")
40 else:
```

9. Output

```
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY> python case10.py
===== DATASET COMPARISON =====

Dataset A: [10, 20, 30, 40, 50]
Dataset B: [28, 29, 30, 31, 32]

===== RESULTS =====
Mean of Dataset A : 30
Mean of Dataset B : 30

Variance of Dataset A : 200
Variance of Dataset B : 2

Standard Deviation of Dataset A : 14.14
Standard Deviation of Dataset B : 1.41

===== INTERPRETATION =====
Dataset A has greater spread.
Dataset B is more consistent.

Mean alone is not enough to compare datasets
because datasets with the same mean can have
different variances and standard deviations.
PS C:\Users\Harshini Darshan\OneDrive\Desktop\DSA0405_JEYAHARSHINIR_192424051\C03\C03_AT1_CASESTUDY>
```

10. Discussion

Question 1: Calculate the Mean

- Dataset A = **30**
- Dataset B = **30**

Both datasets have the same average.

Question 2: Calculate the Variance

- Dataset A = **200**
- Dataset B = **2**

Dataset A has a much larger variance.

Question 3: Calculate the Standard Deviation

- Dataset A = **14.14**
- Dataset B = **1.41**

Dataset B has a much smaller standard deviation.

Question 4: Which Dataset Has Greater Spread?

Dataset **A** has greater spread because its values are distributed much farther from the mean than Dataset B.

Question 5: Which Model Appears More Consistent?

If both datasets represent model errors, **Dataset B** is more consistent because its errors are closely clustered around the mean, resulting in a much smaller variance and standard deviation.

Question 6: Why Is Mean Alone Not Enough?

Although both datasets have the same mean of **30**, they differ significantly in how their values are distributed. Dataset A contains values spread widely from **10 to 50**, whereas Dataset B is tightly clustered between **28 and 32**. Therefore, the mean alone cannot describe the consistency or variability of a dataset. Variance and standard deviation provide additional information about the spread of the data.

11. Interpretation

This analysis demonstrates that datasets with identical averages can behave very differently. Dataset A shows high variability, while Dataset B exhibits very little variation. Therefore, measures of dispersion such as variance and standard deviation are essential for accurately comparing datasets.

12. Real-World Applications

- Machine Learning model evaluation
 - Data Analytics
 - Quality Control
 - Financial Risk Analysis
 - Scientific Research
 - Manufacturing
 - Healthcare Analytics
 - Business Intelligence
-

13. Advantages

- Compares dataset consistency.
 - Measures variability accurately.
 - Supports informed decision-making.
 - Identifies reliable datasets.
 - Useful in predictive modeling.
-

14. Limitations

- Mean does not describe variability.
 - Variance is affected by extreme values.
 - Small datasets may not represent larger populations.
 - Statistical measures alone cannot explain the reasons for variability.
-

15. Conclusion

Both datasets have the same **mean value of 30**, but they differ significantly in their variability. **Dataset A** has a variance of **200** and a standard deviation of **14.14**, indicating a much larger spread of values. **Dataset B** has a variance of **2** and a standard deviation of **1.41**, showing that its values are closely clustered around the mean. Therefore, **Dataset B is more consistent** and would be preferred if the datasets represented model errors. This case study highlights that **mean alone is not sufficient for comparing datasets**, and measures of dispersion are essential for understanding the reliability and consistency of data.